



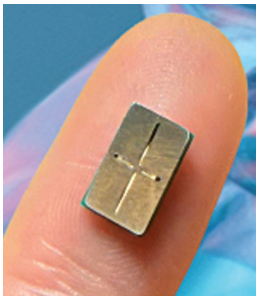
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

World's smallest active micro fan

xMEMS Labs has introduced the XMC-1200, a micro fan designed to cool AR, XR, and smart glasses in spaces such



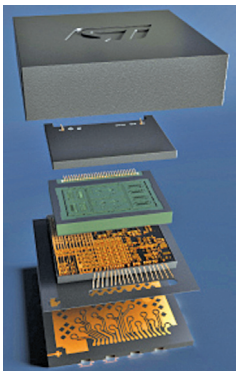
as temple arms where conventional fans cannot fit. As these devices handle more processing, cameras, displays, and on-device

AI, the fan provides cooling at the heat source. In AR and XR devices, it helps control the temperature of microdisplay LEDs and laser light sources, reducing heat-related wavelength shifts that can affect colour, contrast, and brightness. It can also cool processors during AI processing, voice assistance, real-time translation, and computer vision, while helping camera-based glasses manage heat from image sensors and video processing for longer recording sessions.

xMEMS
<https://xmems.com>

Vibration sensor brings high-frequency sensing

STMicroelectronics has introduced the IIS3DWB10IS, a digital vibration sensor for industrial condition monitoring.



It measures vibrations and shocks up to 200g at frequencies above 10kHz and operates at temperatures up to 125°C. Based on ST's

MEMS technology, it integrates the ISPU 2.0 (intelligent sensor processing unit) for signal processing and AI inference, allowing vibration analysis near the sensing element without relying on an external processor. The sensor is designed for rotating and oscillating machinery used in manufacturing, automotive production, and other industrial applications. It can detect vibration changes that may indicate problems such as bearing failures. The ISPU 2.0 delivers up to 40MIPS and 40MFLOPS, four times the previous generation, while its sensor interface supports six times faster data transfer with the MEMS sensing circuitry.

STMicroelectronics
<https://www.st.com/content>

AI chip stacks memory directly on compute layer

China-based startup Dongfang Suanxin (DFSX) has launched the DF1000, an AI chip developed using a domestic semiconductor supply chain.



Built on a 14nm process, the chip combines the compute layer with memory placed directly above it, increasing memory bandwidth and reducing the distance between

memory and processing units without using HBM. DFSX says the DF1000 can deliver performance comparable to some Western AI chips for selected inference workloads, but it is not designed for high-end AI training. The company expects the upcoming DF2000 to improve training

performance. The DF1000 also uses reconfigurable hardware that can adapt to software workloads.

Dongfang Suanxin (DFSX)
<https://www.dongfang.com>

Single-chip automotive radio and audio solution

NXP Semiconductors has introduced the SAF9800, an automotive audio and radio processor that combines AM/FM radio reception, audio processing, and AI/ML



capabilities in a single chip. The device is designed to reduce component count in infotainment systems and support software-defined vehicle architectures. It integrates analogue radio reception with audio processing, allowing automakers to use the same hardware platform across vehicle models and markets. NXP claims more than 10x the audio processing performance of previous generations, supporting noise reduction, voice and sound classification, in-cabin audio processing, and acoustic source separation. The chip also allows audio and radio functions to be configured and updated through software without replacing the hardware.

NXP Semiconductors
<https://www.nxp.com>

Device brings safety alerts to e-two-wheelers

Autotalks has launched a V2X product line for e-bikes and e-scooters that connects vulnerable road users (VRUs) to the V2X safety network. The line includes the TEKTON3-BIKE, a system-on-chip (SoC) based on the TEKTON3-AUTO chipset and designed for